

Lumateperax Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-825/B

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

The control strategy links critical quality attributes to critical process parameters identified during development. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. No changes to the approved analytical procedures are proposed in this submission. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

3.2.P.4 Control of Excipients

For the commercial formulation F-825/B, Ac-Di-Sol SD-711 (Croscarmellose sodium, disintegrant) is used.

Grade selection and control follow the principles of ICH Q6A and ICH Q3D.

Formulation Development Rationale

The formulation development rationale for Ac-Di-Sol SD-711 rests on consistent dissolution performance across registration batches.

Current Commercial Specification

Component	Function	Grade per current specification
Croscarmellose sodium	disintegrant	Solutab Type A
Reference		SPEC-EX-3385 Rev 02

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Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Deviations observed during the campaign were investigated and closed with no impact to product quality. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Deviations observed during the campaign were investigated and closed with no impact to product quality. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

No changes to the approved analytical procedures are proposed in this submission. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits.